TRUNKBELAY · PROBLEM STATEMENT

TrunkBelay problem statement

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	TKB-PRB-001	Version	0.4
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/trunkbelay		
Source	docs/01-problem.md @ b6393c7		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-25	Amish Chadha	Kit 1.9.1 upgrade; users-and-needs and stakeholders tables added (STANDARDS section 22), the source of TKB-URS-001; cost figure updated to USD 828 (TKB-DEC-001 D-S1)
0.3	2026-10-03	Amish Chadha	Cost and carry constraints updated for Amish's decisions 18C and 19A (TKB-DDR-003)
0.2	2026-10-03	Amish Chadha	TRL 2 and 3 update; constraints restated with the value-engineering target, open questions answered, first co-design candidates, safety section
0.1	2026-09-30	Amish Chadha	Initial scaffold

TrunkBelay problem statement

When a climber is stranded on a palm, the first person able to help is usually another climber, not a fire crew. That person can reach the victim but has nothing to anchor them to and no way to lower them.

The problem

The reported rescues follow a pattern. A climber blacks out or slips, the climbing device holds the feet, and the person hangs head down. A bystander or fellow climber goes up and improvises with cloth and rope to stop a fall (Onmanorama, 2025; Onmanorama, 2026). Fire crews then arrive, use ladders and rescue nets, and in some cases an officer climbs the trunk directly (Onmanorama, 2026). Each rescue is improvised on the spot.

Utility workers have purpose-built answers. The Buckingham SuperSqueeze is a pole-top rescue strap with an adjustable web grab, rated to 350 lb and sold at about USD 1,576 (Buckingham). The 3M DBI-SALA Cynch-Lok grips a wooden pole with a cleated plate and carabiner and is rated to 310 lb for poles from 5.5 to 18.5 in diameter (Saftgard). These are sized for smooth utility poles, priced for utilities and not designed for the fibrous, tapering and often curved trunk of a coconut palm.

Users and context

User	Need	Context
Coconut climbers (the rescuer)	Anchor and lower a stranded colleague quickly and safely	Plantations and home gardens, often working in pairs or small groups
The stranded climber	Be supported, freed from the climbing device and lowered without a fall	Head down or slumped, possibly unconscious, 6 to 25 m up
Fire and rescue services	A light anchor to secure a victim before ladders or nets are set	Rural stations called to palm rescues
Coconut Development Board and climber training schemes	A standard kit and drill to teach	Climber training and insurance programmes

Users and needs

Table 2. Users and needs (the source of the user requirements, TKB-URS-001).

User type	Goals and needs	Typical use scenario	Pain points and challenges
Rescuer: a fellow coconut climber	Anchor the stranded climber to the trunk quickly and lower them safely without help from below; keep both hands free while climbing; trust the anchor on a wet trunk	Climbs on a mechanical climbing device to just below the victim, 6 to 25 m up, often within minutes of the event	Nothing to anchor to but the trunk; improvised cloth and rope; working one-handed at height; fear of dropping the victim; the kit must be light and simple enough to carry and remember
The stranded climber	Be held so they cannot fall, turned head up, freed from the climbing device and brought down as soon as possible	Hangs head down or slumped from the device, possibly unconscious	Reported waits of 30 to 150 minutes head down; cannot help in their own rescue
Helpers on the ground: family, other pluckers, neighbours	Pass up what the rescuer needs and take the person at the bottom	Stand at the foot of the palm; call emergency services	Cannot reach the victim; no role without a way to send gear up
Fire and rescue crew	Secure the victim fast on arrival, before ladders or nets are set, and finish the rescue	Arrive by road from a rural station, sometimes after an hour or more	Ladders do not reach tall palms; each rescue improvised on the spot
Trainer in a climber training scheme	A standard kit and a drill short enough to teach in a half day, and kits a scheme can afford	Teaches machine climbing under the Coconut Development Board's Friends of Coconut Tree scheme	No rescue drill or kit in current training; budgets per climber are small

Stakeholders

Table 3. Stakeholders.

Stakeholder group	Role or interest	Engagement needed
Coconut climbers and their collectives or labour banks	First users and rescuers; fit trials on real palms	Co-design of the kit and drill; fit and carry trials (TRL 4)
Kerala Fire and Rescue Services stations with palm rescues (Kasaragod, Kannur)	Respond to stranded climbers; could adopt the kit as a quick anchor	Review of the drill; joint dummy drill at low height
Coconut Development Board and its Friends of Coconut Tree training scheme	Trains climbers; runs the climbers' insurance scheme	Training trial (R10); group purchase of the rescue gear
Palm owners and farmers	Own the palms; care that the trunk is not damaged	Consent for trials on their palms; inspection of bark after use (R8)
Local fabricators with an AC TIG welder	Make and repair the aluminium frame	Quotes, a sample frame and weld inspection (TRL 4)
Rope rescue equipment suppliers	Supply the certified descender, carabiners, rope and triangle	Data on the bought parts (descender hand force, rope stretch)
Design Molecule	Publishes the design as open hardware	Keeps it open and clear of the Cynch-Lok cinch geometry

Operating environment

- Coconut palm trunks roughly 200 to 450 mm (8 to 18 in) diameter (estimate, to be surveyed), fibrous, often wet, sometimes curved or leaning.
- Working heights from about 6 m to 25 m (20 to 82 ft).
- Hot humid climate, monsoon rain; kit stored in the open or in a shed.
- Rescuer is already on the trunk using a mechanical climbing device or traditional rope loop.
- No anchor points other than the trunk itself.

Constraints

- Value-engineering target: USD 900 for the first prototype kit (a control target for value engineering, not a spending limit; STANDARDS section 18).
- Kit parts cost target under USD 850 per kit at single-kit prices (R9), restated by Amish on 2026-10-03 from USD 100 so that the auto-locking descender and the evacuation triangle are kept; group purchase by the training partner and one kit per climber group are to bring the cost per climber down.
- Carried up the trunk by one climber: under 5 kg (11 lb) (target); the rope bag may be hauled up by the helpers on a tag line (decision 18C).
- Collar must fit the full trunk range without special tools.
- Must not use the ratcheting cinch geometry of the 3M Cynch-Lok (IP screen design-around).
- Open hardware under CERN-OHL-S-2.0.

Out of scope

- Replacing or redesigning climbing devices.
- Fall arrest for routine climbing (a separate problem).
- Rescue of a climber who has already fallen.
- Medical care beyond getting the person to the ground.

Prior work

Prior work	What it does	Gap for these users	Source
Buckingham SuperSqueeze pole-top rescue (488P)	Adjustable web-grab strap for lowering an injured lineman from a utility pole	Priced for utilities (about USD 1,576) and sized for smooth poles, not fibrous palm trunks	link
3M DBI-SALA Cynch-Lok pole climbing device	Fall-restriction strap gripping the pole with a cleated plate and carabiner	Fall restriction for the climber, not a rescue anchor; proprietary cinch geometry	link
CA2681870C, Bashlin pole climbing and fall restraint device (lapsed 2015)	Belt and lanyard with a locking device for pole climbing	Personal restraint, not a rescue anchor	link
WO2012131439A1, safety tree climbing device (ceased)	Engine-powered wheeled drum that climbs single-stem palms with a brake and safety cable	Climbing machine, not a rescue kit; never entered national phase	link
Fire service improvised rescue with ladders, ropes and nets	Current practice in Kerala palm rescues	Arrives late; relies on ladders that do not reach tall palms	link

Co-design

A climber training programme or climbers' collective in Kerala, working with a local Fire and Rescue Services station that has done palm rescues, so the kit and drill fit both the climbers who will use it first and the crews who arrive later.

First candidates to approach (none approached yet; recorded in TKB-DDR-001, D8):

1. A coconut climber training programme run under the Coconut Development Board's "Friends of Coconut Tree" scheme in Kerala, for the drill and the training trial (R10).
2. A Kerala Fire and Rescue Services station in a district with recorded palm rescues (Kasaragod or Kannur), for the drill and for adoption by crews.
3. A climbers' collective or labour bank in the same district, for fit trials on real palms (R3) and the kit's carry and rigging (R4, R7).

Questions answered at TRL 2 and 3

The open questions of version 0.1 were settled on paper under Amish's pre-approval of 2026-10-03 (TKB-DDR-001). Each answer is checked at TRL 4.

Question (version 0.1)	Answer on paper	Record
Can the collar hold the load turning from upward to downward as the victim passes it?	Not with the single top chain. The drill sets the collar at least 300 mm above the point where the rope meets the victim, so the pull is always downward. Whether that drill is workable is a decision for Amish (R2)	TKB-DDR-001, D3; TKB-DEC-001
How much does grip vary with moisture, age and curvature?	The collar is self-locking: it needs a combined friction of 0.58 or less on a 200 mm trunk against about 0.9 assumed for rubber and sleeved chain on wet bark. Friction on real trunks is the first thing to confirm	TKB-CAL-001, section B
Which friction device is simplest to learn?	An auto-locking descender with an anti-panic function, plus a carabiner the brake strand turns over. It stops if the rescuer lets go	TKB-DDR-001, D4
How is an unconscious, head-down person fitted with a sling?	An evacuation triangle round the hips and a chest sling, both clipped to the rope's victim carabiner, so the person turns head up when the feet are freed	TKB-DDR-001, D5
Would fire services adopt the kit, and would that change the rating?	Unknown until the first co-design candidates are approached. The rating stays at one person (100 kg design mass, 2.5 kN static)	TKB-DDR-001, D8

Safety

Safety: TrunkBelay is a rope rescue and work-at-height kit. A mistake can drop a person from 6 to 25 m. It is published as an open engineering reference, not certified rescue or fall-protection equipment, and is not used on a person before a proof test of the collar on cut trunk sections and a supervised drill at low height (TRL 4). Emergency services are always called as well. A person who has hung head down for any length of time needs medical assessment after the rescue.